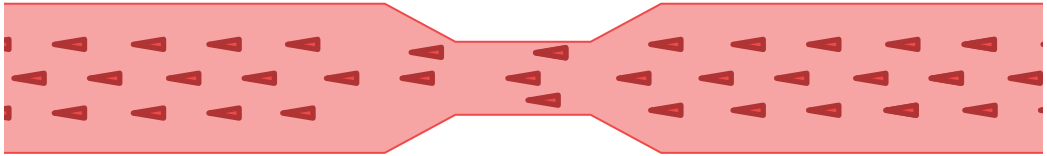
FLOW TRHOUG VENTURIMETER

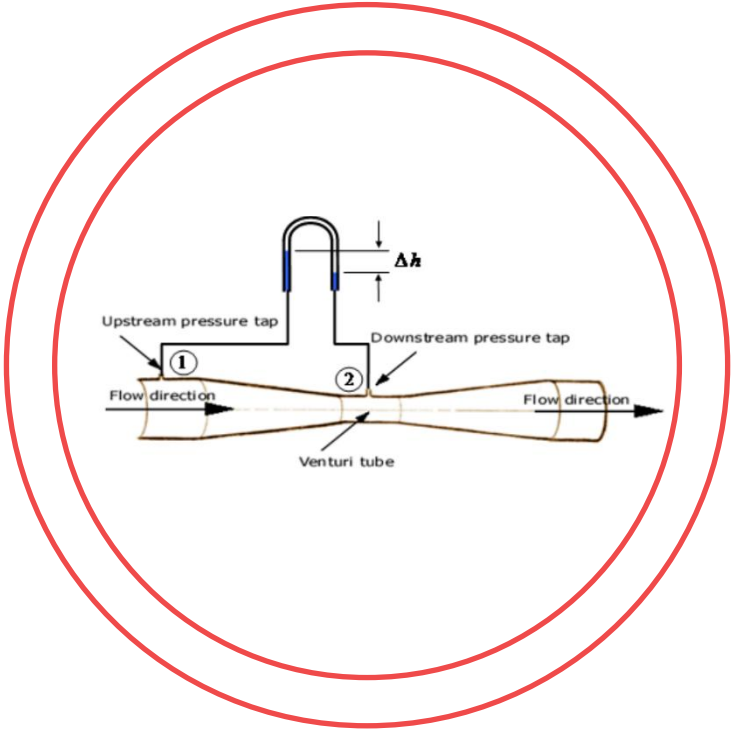


Our Team Layout

Anika Tahsin	Arijit Shaha	Shahlan Kabir	Tahsin Ahmed	Mehedi Hasan
Introduction	Procedure	Analysis	Discussion	Journey

Introduction

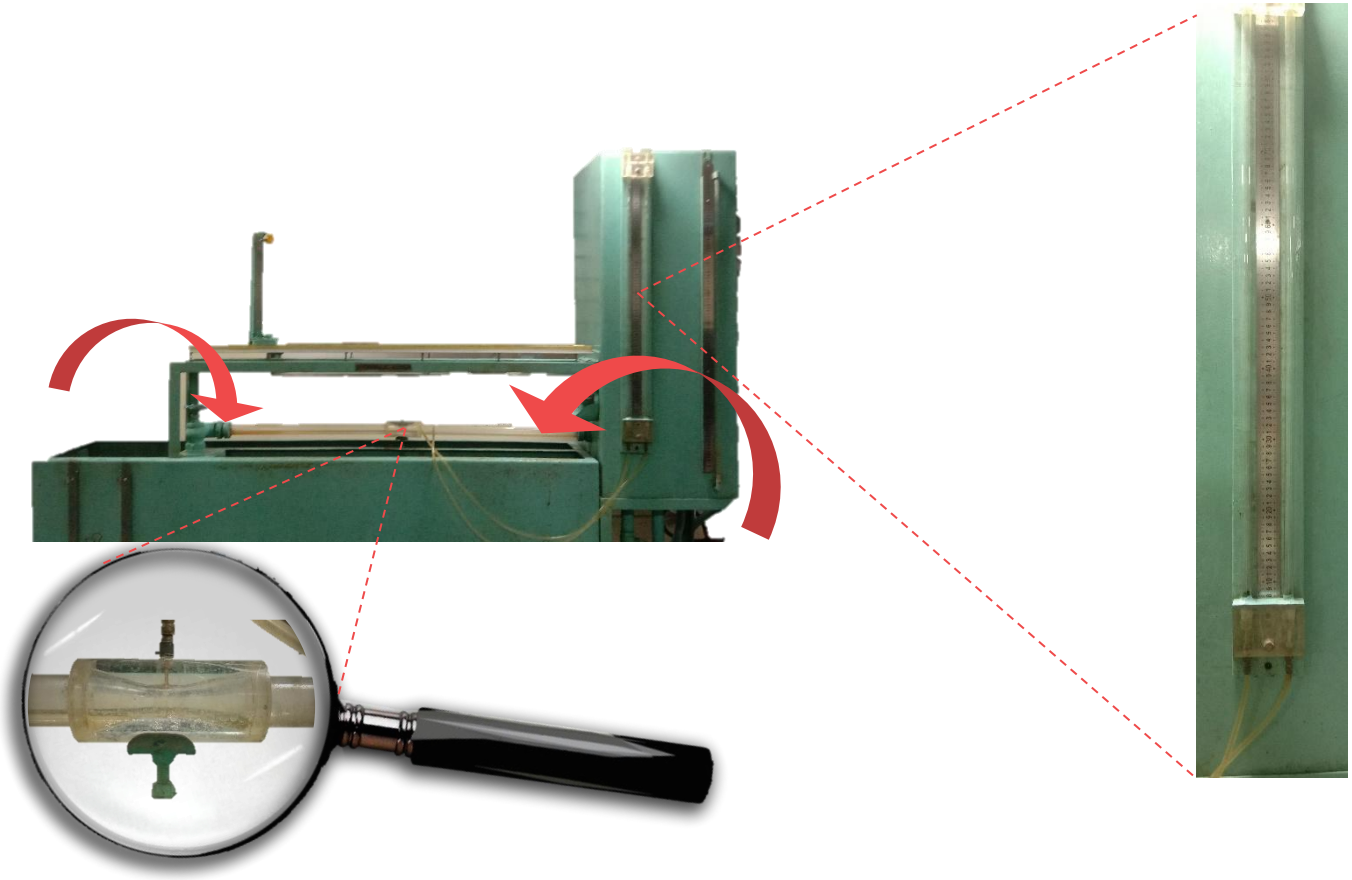
- Considered as a meter device to measure the flow
- To increase the velocity
- Works on the principle of Bernoulli's theorem
- The use of low pressure and high velocity



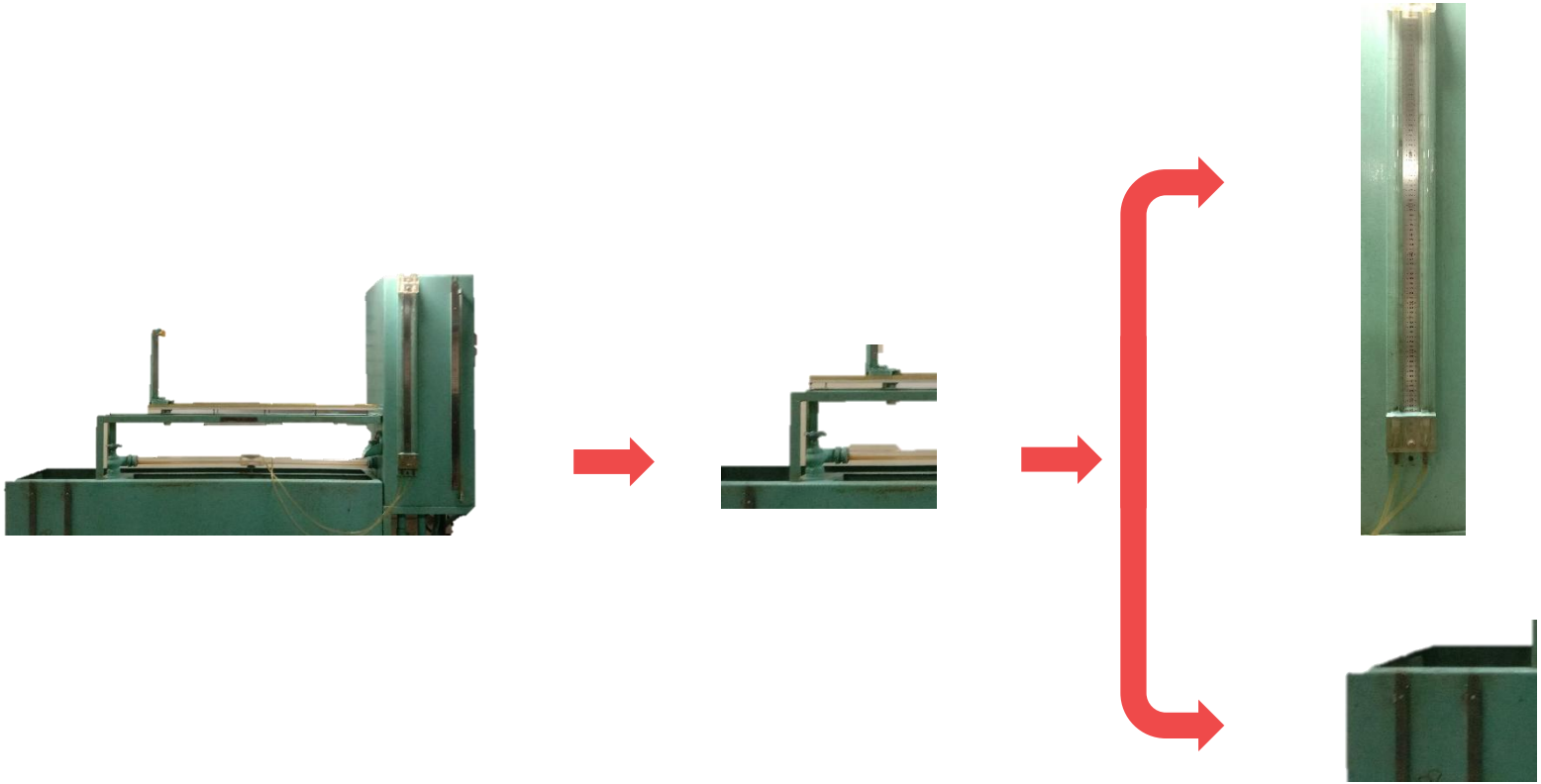
Objective

- To obtain the co-efficient of discharge
- Relation between Reynolds number.
- To compare the value between theoretical and experiment al discharge.

Procedure



Data Collection



Analysis

Mean and Median

Mean value of Cd

0.936835856

Median value of Cd

0.949232887

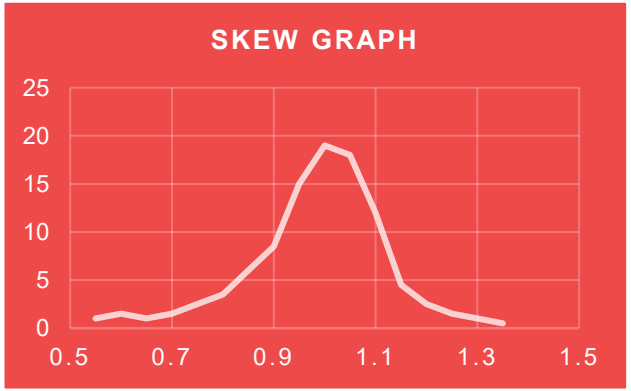
Standard value of Cd

0.95 ~ 0.99

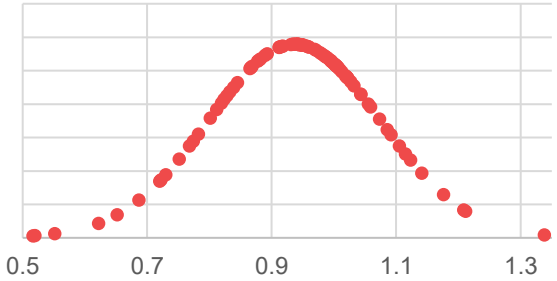
Analysis

Skewness and Standard deviation

Skewness (-0.64876726)



Standard Deviation

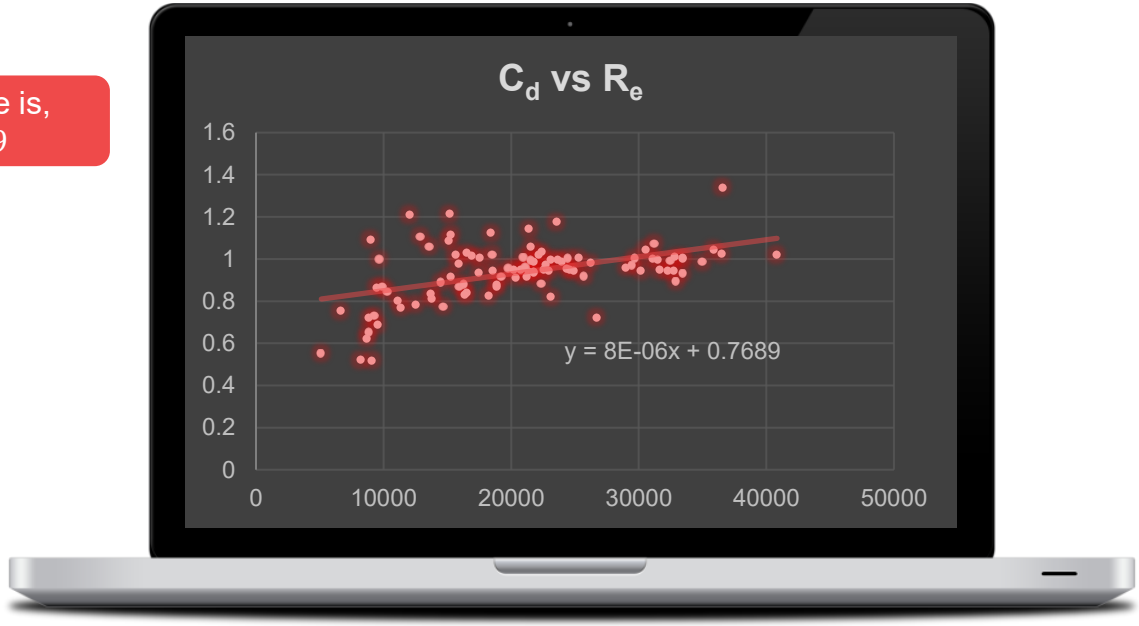


Analysis

Coefficient of Discharge VS Reynolds Number

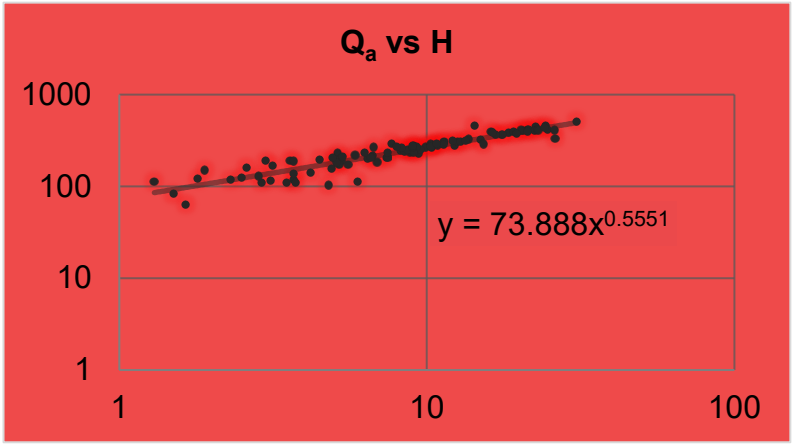
01 It forms a straight line.

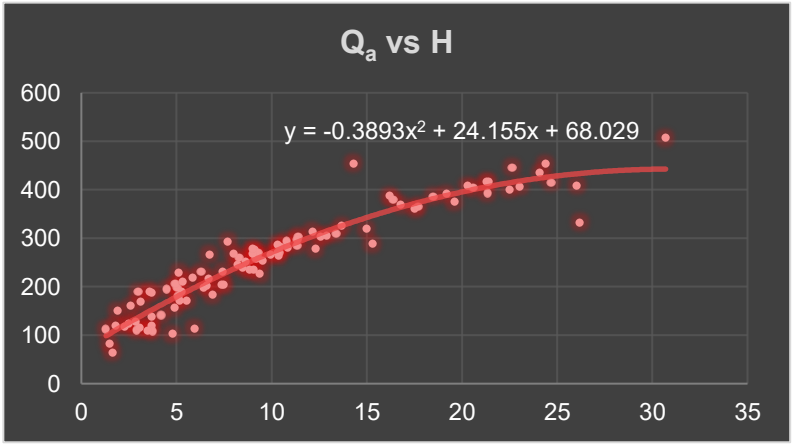
02 The equation of the straight line is,
$C_d = R_e * 8 * 10^{-06} + 0.7689$



Analysis

Q_a VS H



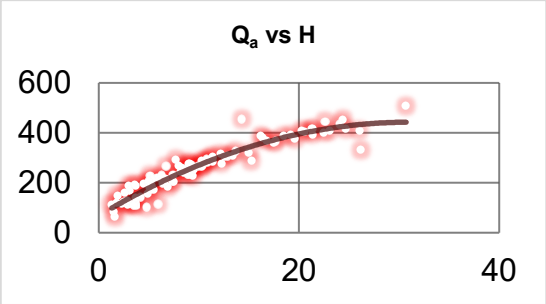


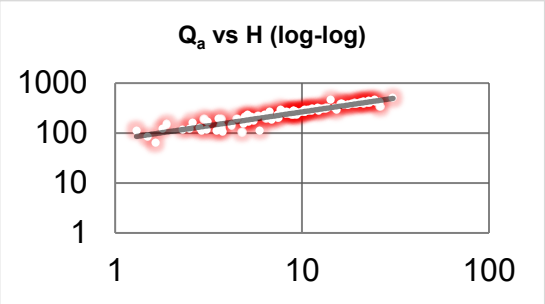
- Forms a straight line
- Calibration Equation is, $Q_a = 73.888H^{0.5551}$

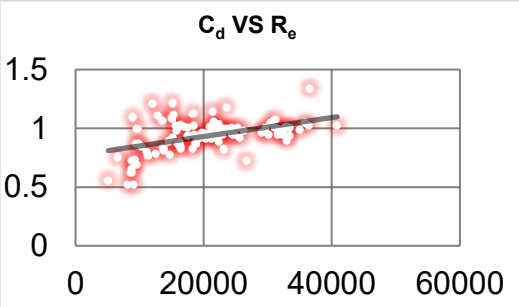
- Forms a curve line,
- Calibration Equation is, $Q_a = -0.3893H^2 + 24.155H + 68.029$

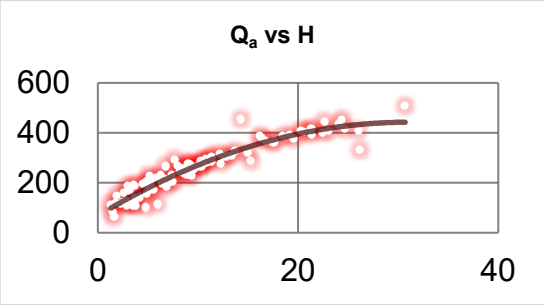
Log- log Graph

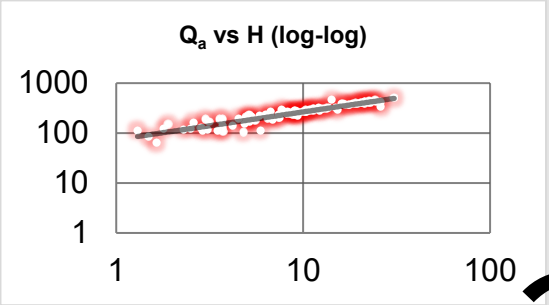
Plain Graph


Q_a vs H
600
400
200
0
0
20
40

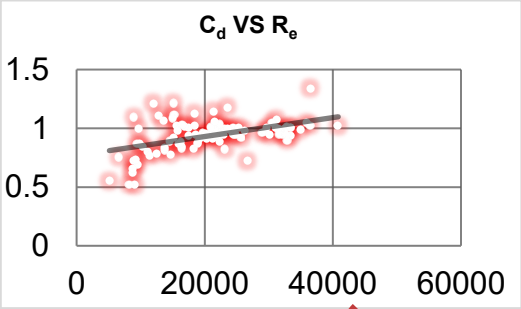

Q_a vs H (log-log)
1000
100
10
1
1
10
100

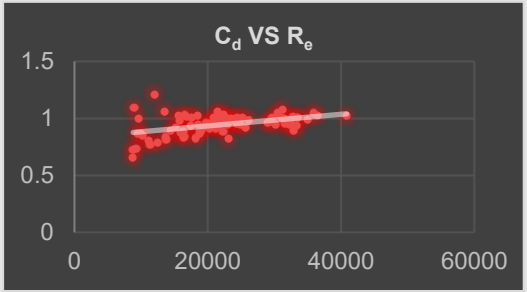

C_d VS R_e
1.5
1
0.5
0
0
20000
40000
60000

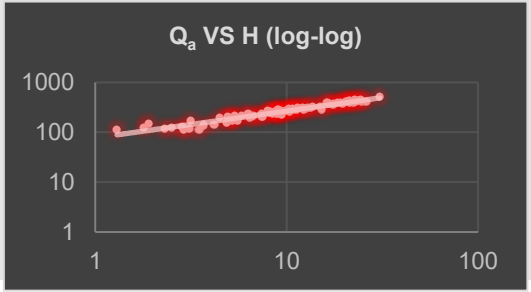


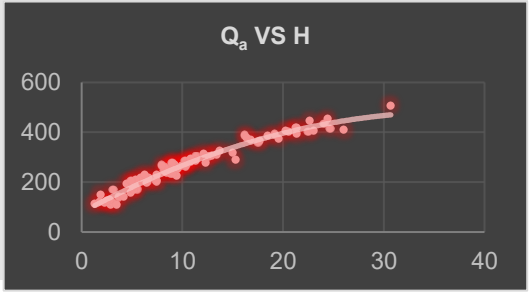


After
Deducting
Less Accurate
Observations









Using
Standard
Deviation

Analysis

Patterns

14 out of 16 inaccurate readings are in between 1 to 8 cm head difference.

Diff H	Acceptibility
2.9	NOT OK
3	NOT OK
3.1	NOT OK
3.15	OK
3.5	NOT OK
3.6	NOT OK
3.7	NOT OK
3.7	NOT OK
3.7	OK
3.75	NOT OK
4.2	NOT OK
4.5	OK
4.8	NOT OK
4.9	NOT OK

Diff H	Acceptibility
20.3	OK
20.6	OK
21.3	OK
21.35	OK
21.4	OK
22.5	OK
22.65	OK
23.05	OK
24.1	OK
24.4	OK
24.7	OK
26.05	OK
26.2	NOT OK
30.7	OK

Acceptability According to the Standard Deviation

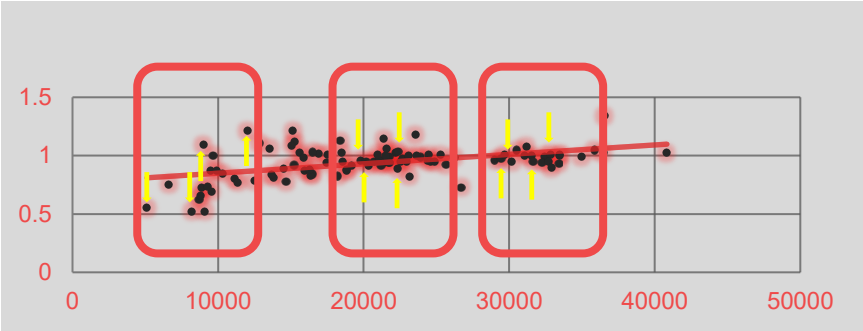
Actual Discharge	
63.52941	NOT OK
82.5	NOT OK
102.1154	NOT OK
108	NOT OK
109.2857	NOT OK
109.9349	NOT OK
111.5702	NOT OK
112.9896	NOT OK
115.1877	NOT OK
117.5958	OK
118.4211	NOT OK

Actual Discharge	
406.2187	OK
407.5472	OK
408.472	OK
414.7465	OK
415.3846	OK
415.3846	OK
433.9286	OK
452.7163	OK
453.629	NOT OK
506.5666	OK

Higher rate of discharge give more accurate result of C_d.

Analysis

Co-efficient of discharge VS Reynolds number



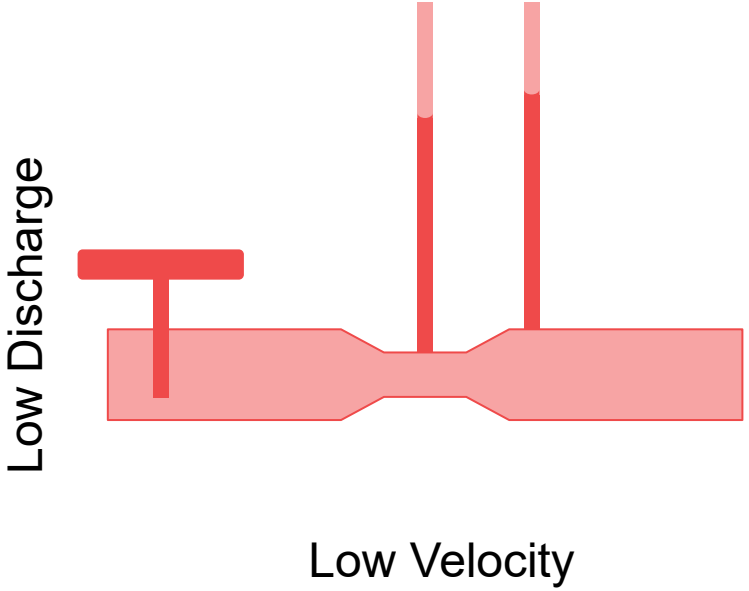
Cd	Re
0.948744	22578.12
0.971869	22698.1
0.943896	22974.49
0.996161	23097.88

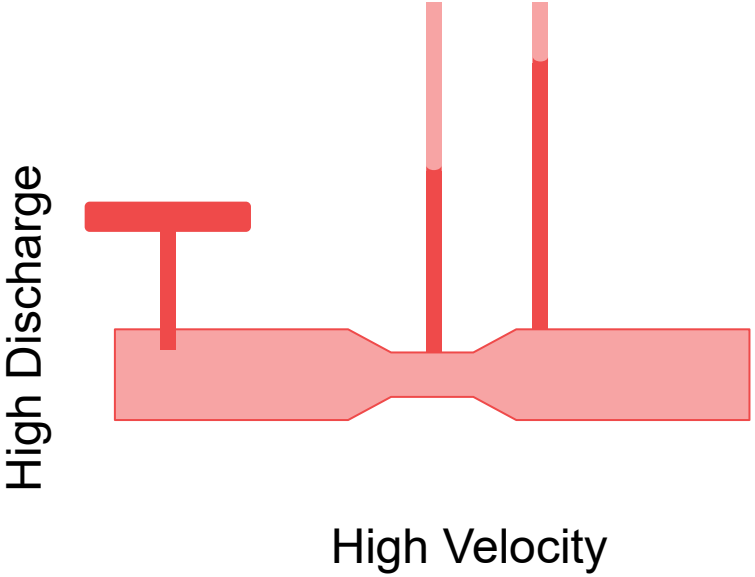
Cd	Re
0.801522	11138.85
0.768407	11377.32
1.208831	12038.33
0.782548	12515.05

- Higher the Reynolds number, the value of C_d is closer to the standard value.

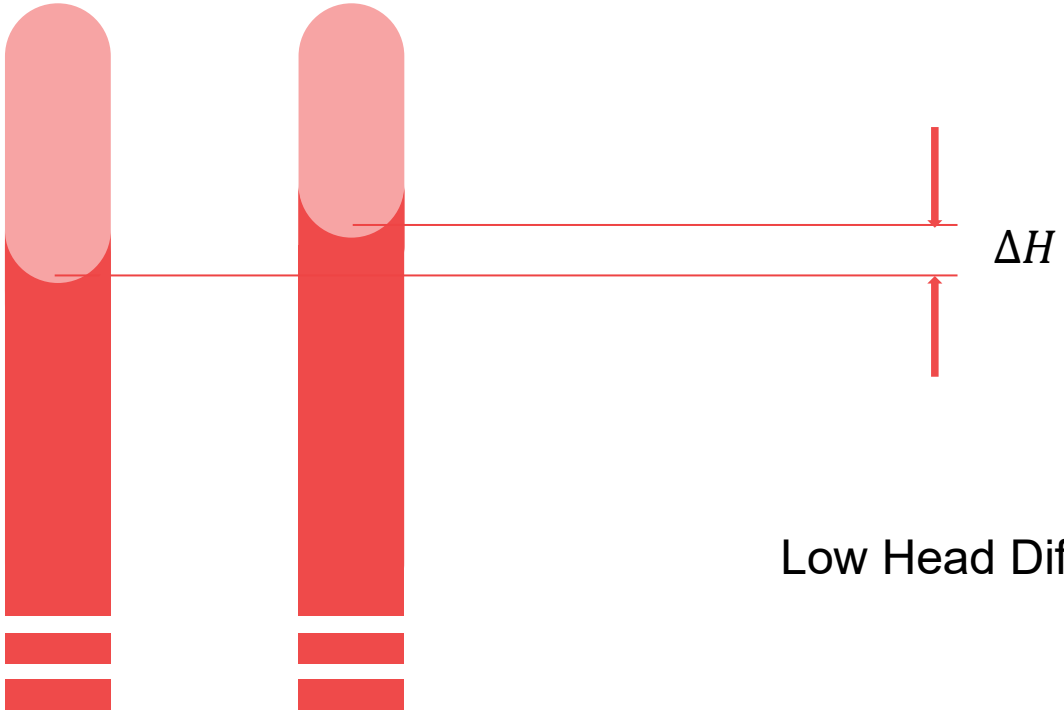
- Lower the Reynolds number, the value of C_d is far from the standard value.

Discussion



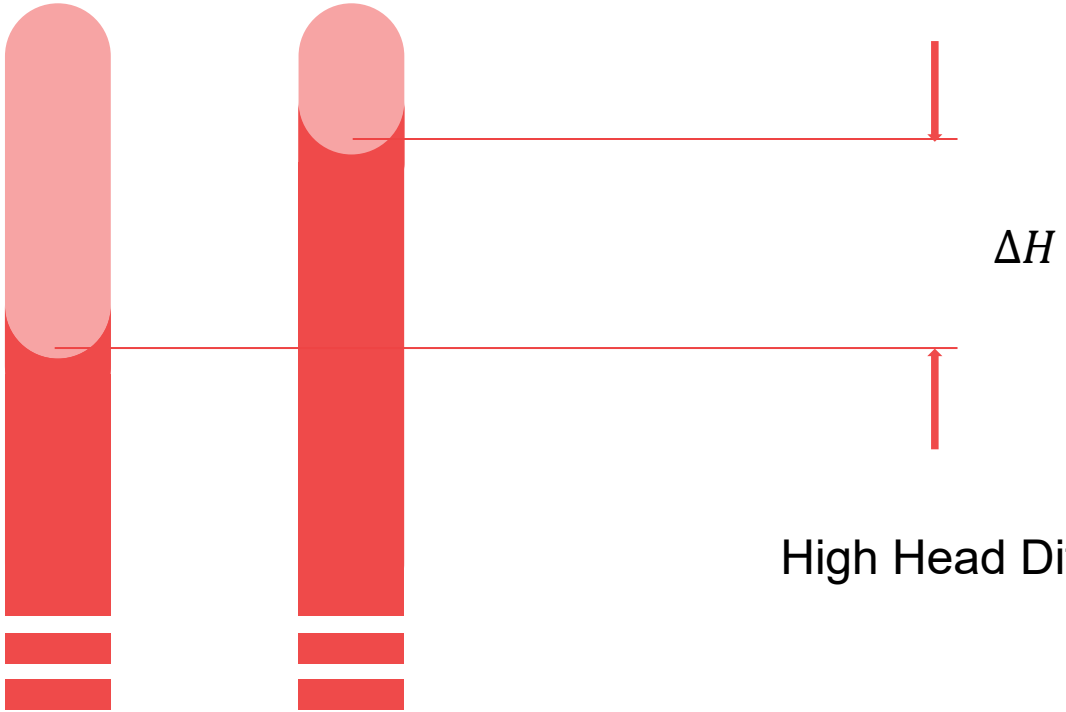


Discussion



Low Head Difference

Discussion



High Head Difference

Recommendation

Smooth surface will provide more accurate C_d by lowering the difference between Q_a and Q_t.

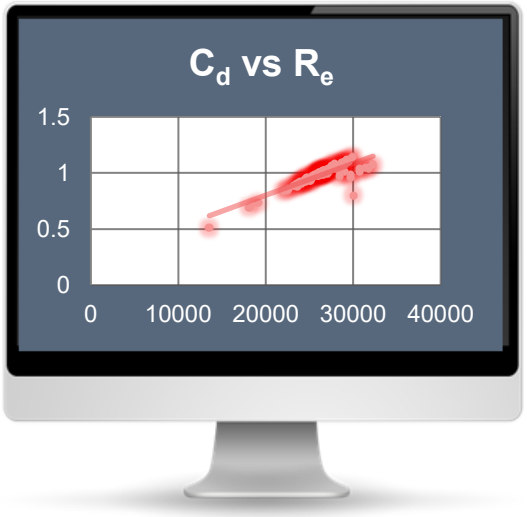
Throat diameter has to decrease for getting standard C_d.

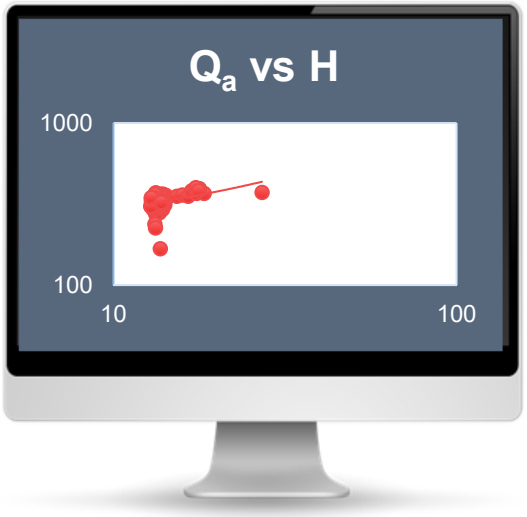
Have to ensure turbulent flow to get better C_d

Project Journey

Journey

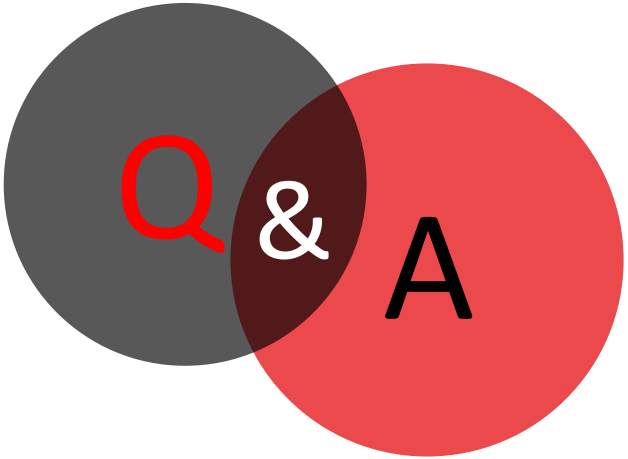
Results from the First observations





Special Thanks

- Abir Hasan & Other



5 MINUTES

THANK You